Tutorial week #8

9 de dezembro de 2025 09:51

1) $C = 20 \quad e = 13 \quad n = 77$

C - ciphertext public key - $\begin{cases} e = 13 \\ n = 77 \end{cases}$ M - message

$M = ?$

We need to compute now the private key d

$d = e^{-1} \pmod{\phi(n)}$

$\phi(n) = \phi(77) = \phi(7)\phi(11) = (7-1)(11-1) = 60$

$\therefore \phi(n) = 60$

$d = 13^{-1} \bmod 60$

d is the value such that $13d \equiv 1 \pmod{60}$

$60 = \underline{13} \times 4 + \underline{8}$

$13 = \underline{8} \times 1 + \underline{5}$

$8 = \underline{5} \times 1 + \underline{3}$

$5 = 3 \times 1 + 2$

$\underline{3 = 2 \times 1 + 1}$ *

* $1 = 3 - 2$

sub. $2 = 5 - 3$

$1 = 3 - (5-3) \iff 1 = 3 \times 2 - 5$

sub. $3 = 8 - 5$

$1 = (8-5) \times 2 - 5 \iff 1 = 2 \times 8 - 5 \times 2 - 5 \iff$

$\iff 1 = 2 \times 8 - 5 \times 3$

sub. $5 = 13 - 8$

$1 = 2 \times 8 - (13-8) \times 3 \iff 1 = 2 \times 8 - 13 \times 3 + 8 \times 3 \iff$

$\iff 1 = 5 \times 8 - 13 \times 3$

sub. $8 = 60 - 13 \times 4$

$1 = 5 \times (60 - 13 \times 4) - 13 \times 3 \iff$

$\iff 1 = 60 \times 5 - 13 \times 20 - 13 \times 3 \iff$

$\iff 1 = 60 \times 5 - 13 \times 23 \iff$

$\iff 1 = -13 \times 23 \iff 1 = 13 \times (-23)$

$13^{-1} \pmod{60} \equiv -23 \equiv 37$

Algoritmo Euclidiano Estendido

$\therefore d = 37$

$c = 20 = m^e \pmod{n}$

$m = c^d \bmod n \iff m = 20^{37} \bmod 77$

Using Python for calculations...

$m = 48$

2) $e = 65 \quad n = 2881 \quad P_s = (n, d)$

$n = pq$, so $43 \times 67 = 2881$

$\phi(n) = (p-1)(q-1) = (43-1)(67-1)$

$= 42 \times 66 = 2772$

$d = e^{-1} \pmod{\phi(n)} \iff$

$\iff d = 65^{-1} \pmod{2772}$

$\iff 65d = 1 \pmod{2772}$

Com Algoritmo de Euclides estendido obtemos

$\iff 65 \cdot 725 \equiv 1 \pmod{2772}$

$P_s = \langle 2881, 725 \rangle$

3) Alice sends to Bob $m^e \pmod{n}$

This is not safe. Because public key is, needless to say, public, it makes PKC vulnerable to a dictionary attack if the message comes from a relatively small set of admissible messages.

Moreover RSA is a multiplicative homomorphism, i.e.

$$E_K(x_1)E_K(x_2) = (x_1^e \pmod{n})(x_2^e \pmod{n})$$
$$= (x_1 x_2)^e \pmod{n} = E_K(x_1 x_2)$$

and this can give origin to some attacks in some contexts. We say that this weakness makes textbook RSA encryption <u>malleable</u>

4) The code was implemented in <u>another file</u>

5) This is <u>NOT</u> safe at all

We know that $d = e^{-1} \pmod{\phi(n)}$, so,

$d \times e = 1 \bmod \phi(n)$, so this means that for some integer K, $d \times e = 1 + K \times \phi(n)$

Then,

$d \times e - 1 = K \times \phi(n) \iff d \times e - 1 = K \times (p-1)(q-1)$

Since d and e are known at this point by the attacker, he can use a probabilistic algorithm (similar to Miller-Rabin) to find the factors p and q.

Once p and q are known, the attacker can compute $\phi(n) = (p-1)(q-1)$ and find the new private key d'.